

COTTON DISEASE DIAGNOSTIC PROTOCOL

1. The Diagnostic Framework

In field pathology, making an accurate diagnosis requires separating biotic (infectious pathogens like fungi, viruses, and bacteria) from abiotic factors (non-infectious environmental issues like salinity, nutrient imbalances, or drought). Agronomists utilize a structured process based on field-level macro patterns down to cellular micro-symptoms before drafting a recommendation.

2. Agronomist Investigative Questionnaire

Category I: Variety & Crop Phenology (The Genetic & Structural Baseline)

- What specific cotton variety did you sow, and is it a Bt or non-Bt variety? (Helps identify genetic resistance or known vulnerabilities to specific regional strains).
- What is the exact age of the crop, or how many Days After Sowing (DAS) is it? (Certain diseases target specific stages, e.g., seedling damping-off vs. mid-season wilts).
- What was the germination percentage, and did you notice any early seedling mortality? (Points toward seed-borne or early soil-borne fungal pathogens).

Category II: Spatial & Chronological Patterns (Macro-Field Level)

- When exactly did you first notice these symptoms in the field? (Establishes speed of disease progression; rapid shifts indicate physiological issues, while slow spreading points to biological pathogens).
- Are the symptoms distributed uniformly across the entire field, or are they appearing in isolated patches or low-lying areas? (Uniform damage points to abiotic stresses; localized patches point to soil-borne fungi, wilt pockets, or nematodes).
- Are the affected plants close to the field edges, water channels, or near weed clusters? (Edge-effects often indicate viral diseases like CLCuV moving into the field via insect vectors hosted on surrounding weeds).

Category III: Symptom Anatomy & Tissue Analysis (Micro-Plant Level)

- Look closely at the leaves: Are they curling upward or downward, and do you see thick green veins or leaf-like growths (enations) underneath? (Classic diagnostic indicators for Cotton Leaf Curl Virus).
- If there are spots on the leaves, what shape are they? Are they rounded with concentric rings, or are they restricted by the leaf veins, looking angular and water-soaked? (Angular water-soaked spots indicate Bacterial Leaf Blight; circular rings suggest fungal spots like Alternaria).
- Where on the plant did the symptoms start—on the older, lower leaves first, or on the fresh, top leaves? (Fungal wilts and mobile nutrient deficiencies show on lower leaves first; viruses and immobile deficiencies show on new growth).
- If you pull up a wilting plant and slice open the main stem vertically, what color is the internal wood/pith? Is it clean white or stained brown/black? (A brown discolored vascular ring confirms a fungal wilt complex like Fusarium or Verticillium; clean white indicates physiological para-wilt).

Category IV: Water, Soil & Root Architecture

- When was your last irrigation, what method did you use (flood vs. drip), and did water stand in the field for more than 24 hours? (Prolonged waterlogging suffocates roots, inducing root rot or physiological drooping).
- What is the typical pH and EC (salinity) of your soil or tube-well water? (High pH >8.0 can lock up micronutrients like iron or zinc, mimicking viral yellowing).
- Is the soil in the affected patch heavily compacted, or is there a hardpan below the surface? (Compaction restricts taproot elongation, causing severe drought/wilt symptoms even if topsoil is damp).

Category V: Crop Nutrition & Chemical Input History

- How many bags of Nitrogen (Urea), Phosphorus (DAP), and Potassium (SOP/MOP) have you applied so far? (Excessive nitrogen creates a lush, succulent canopy highly vulnerable to sucking pests and blight; potassium deficiency causes leaf margin scorch).
- Have you sprayed any weedicides (herbicides) recently? If yes, what was the chemical, the dose, and did you use a completely washed tank? (Herbicide drift or residue causes leaf curling, cupping, and twisting frequently mistaken for virus damage).

Category VI: Environmental Microclimate & Pest Vectors

- What has the weather been like over the past 7 to 10 days? Has it been hot and dry, or cloudy with intermittent rain and high humidity? (High humidity with splashing rain drives fungal/bacterial spore dispersal; hot, dry spells trigger insect vector surges).
- What is the current population of sucking pests, specifically whiteflies, jassids, or thrips, in your field? (Whiteflies are vectors for CLCuV; heavy jassid feeding causes hopper burn which mimics fungal leaf damage).

Category VII: Historical Cultivation & Previous Actions

- What crop was grown in this exact field during the last winter (Rabi) season? (Certain crops host identical soil fungi or nematodes, compounding disease pressure).
- Did you treat the cotton seeds with a fungicide/insecticide protectant before sowing? (Seed treatments delay or prevent early-season vascular wilts and seedling blights).
- Have you already sprayed any fungicide or bio-stimulant to cure this specific problem, and did the plant show any recovery after it? (Prevents chemical duplication and avoids phytotoxicity from over-spraying).